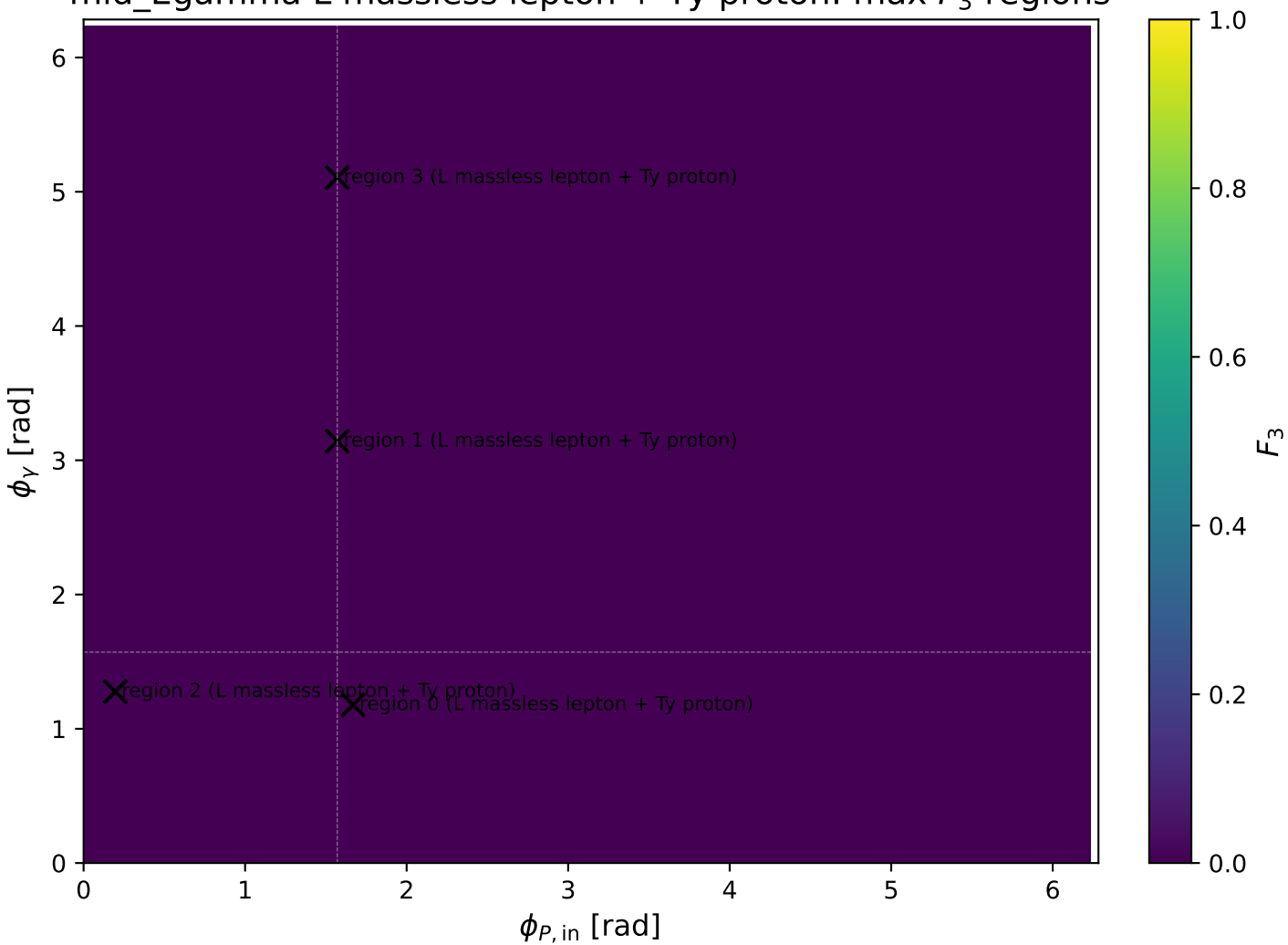
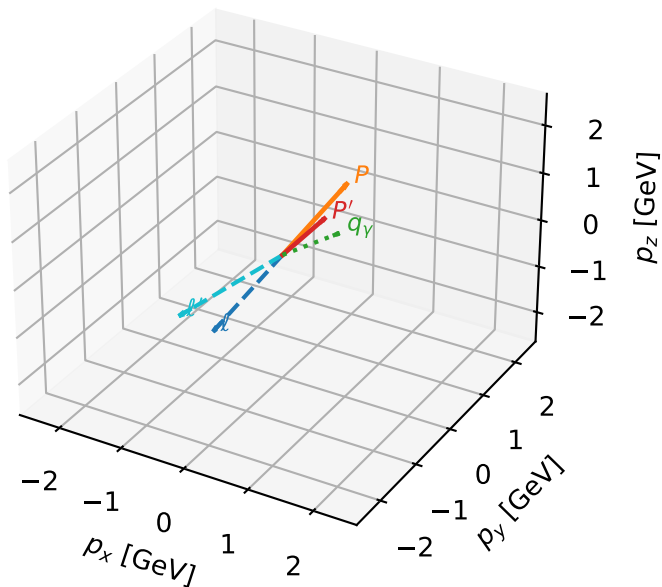


mid_Egamma L massless lepton + Ty proton: max F_3 regions



L massless lepton + Ty proton characteristic max F_3 configuration

3D momenta

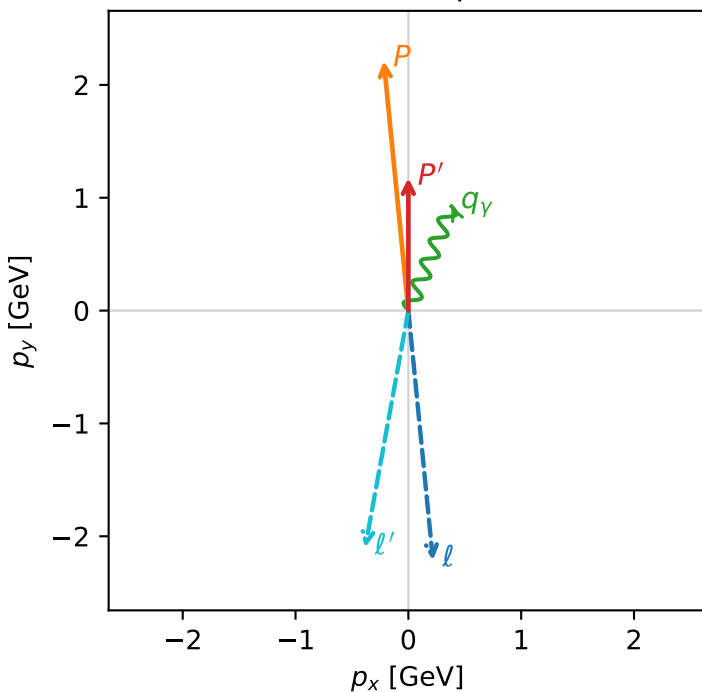


f3_mid_egamma_lty_region_0 $F_3=3.40866\text{e-}08$
 region: mid_Egamma_LTy_region_0
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|L_{\ell_0}\rangle \otimes |T_{y,p}\rangle$

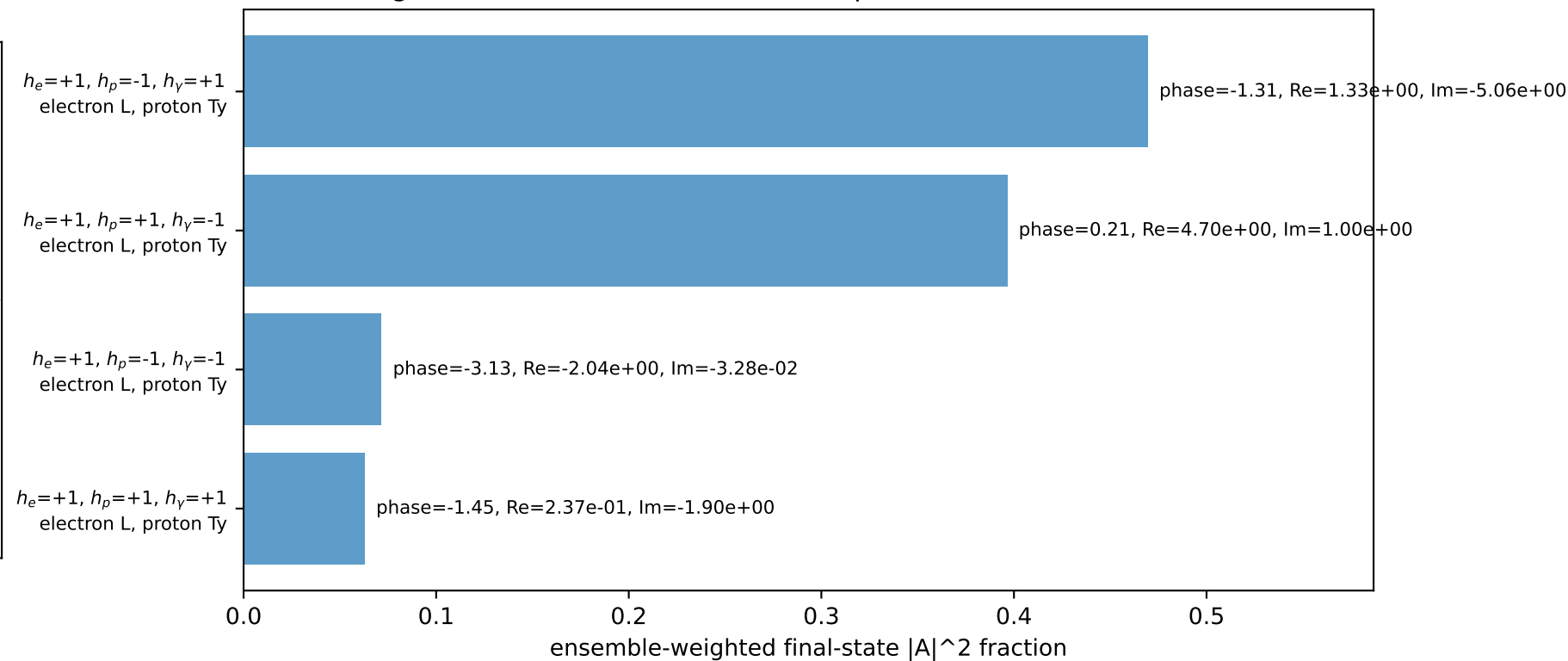
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=1.17588$
 $\theta_{in}=1.57079$, $\phi_l=4.81056$, $\phi_P=1.66897$
 $E_Y=1$, $\phi_Y=1.1781$
 $Q^2=0.365729$, $x_B=0.304436$, $t=-0.296656$, $W^2=1.71545$, $y=0.0582154$
 $\theta(\ell', \gamma)=167.829$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= 0.2180$ $p_y= -2.2137$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= -0.2180$ $p_y= 2.2137$ $p_z= 0.0000$
 ℓ' $E= 2.1343$ $p_x= -0.3827$ $p_y= -2.0998$ $p_z= 0.0000$
 P' $E= 1.5042$ $p_x= 0.0000$ $p_y= 1.1759$ $p_z= 0.0000$
 q_Y $E= 1.0000$ $p_x= 0.3827$ $p_y= 0.9239$ $p_z= 0.0000$

Transverse plane

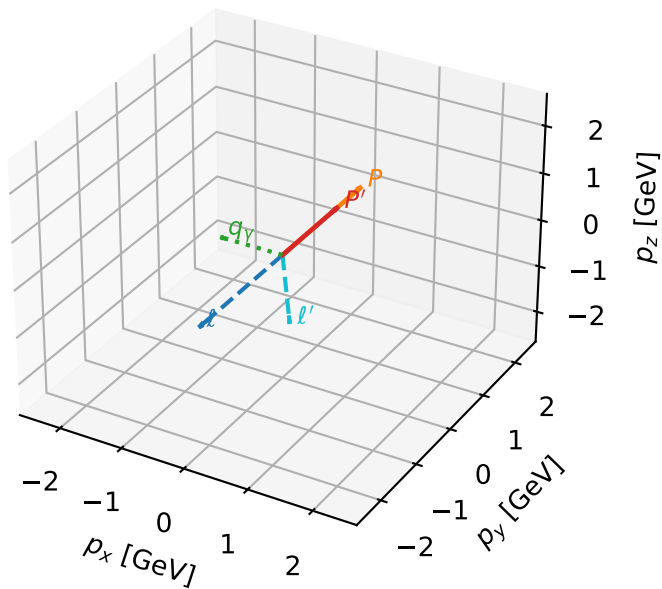


Leading initial-ensemble/final-state components (N=4, retained=100.0%)



L massless lepton + Ty proton characteristic max F_3 configuration

3D momenta



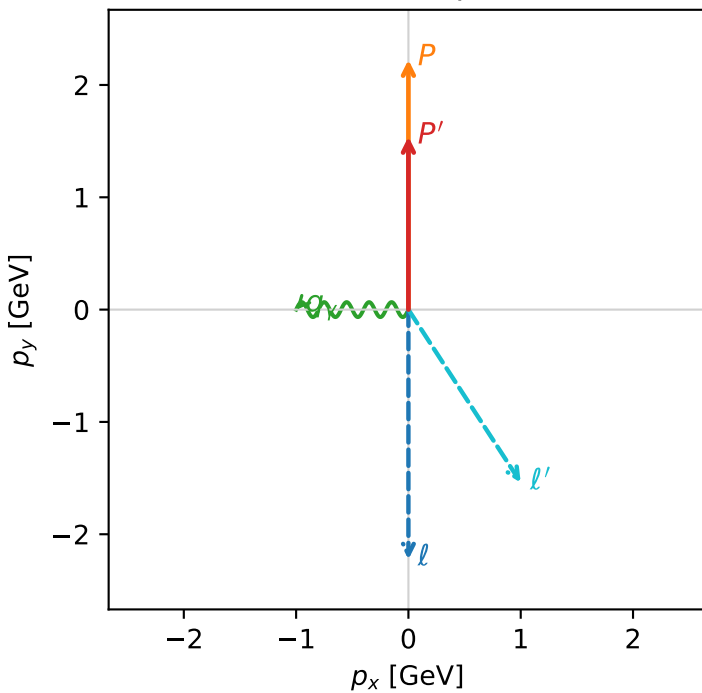
f3_mid_egamma_lty_region_1 F_3=3.29632e-08
 region: mid_Egamma_LTy_region_1
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|L_{\ell_0}\rangle \otimes |T_{Y,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=1.5395$
 $\theta_{in}=1.57079$, $\phi_{\ell}=4.71239$, $\phi_P=1.5708$
 $E_{\gamma}=1$, $\phi_{\gamma}=3.14159$
 $Q^2=1.31807$, $x_B=0.267706$, $t=-0.0953632$, $W^2=4.48534$, $y=0.238591$
 $\theta(\ell', \gamma)=123.006$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 2.2244$ $p_x= -0.0000$ $p_y= -2.2244$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= 0.0000$ $p_y= 2.2244$ $p_z= 0.0000$
 ℓ' $E= 1.8358$ $p_x= 1.0000$ $p_y= -1.5395$ $p_z= 0.0000$
 P' $E= 1.8027$ $p_x= 0.0000$ $p_y= 1.5395$ $p_z= 0.0000$
 q_{γ} $E= 1.0000$ $p_x= -1.0000$ $p_y= 0.0000$ $p_z= 0.0000$

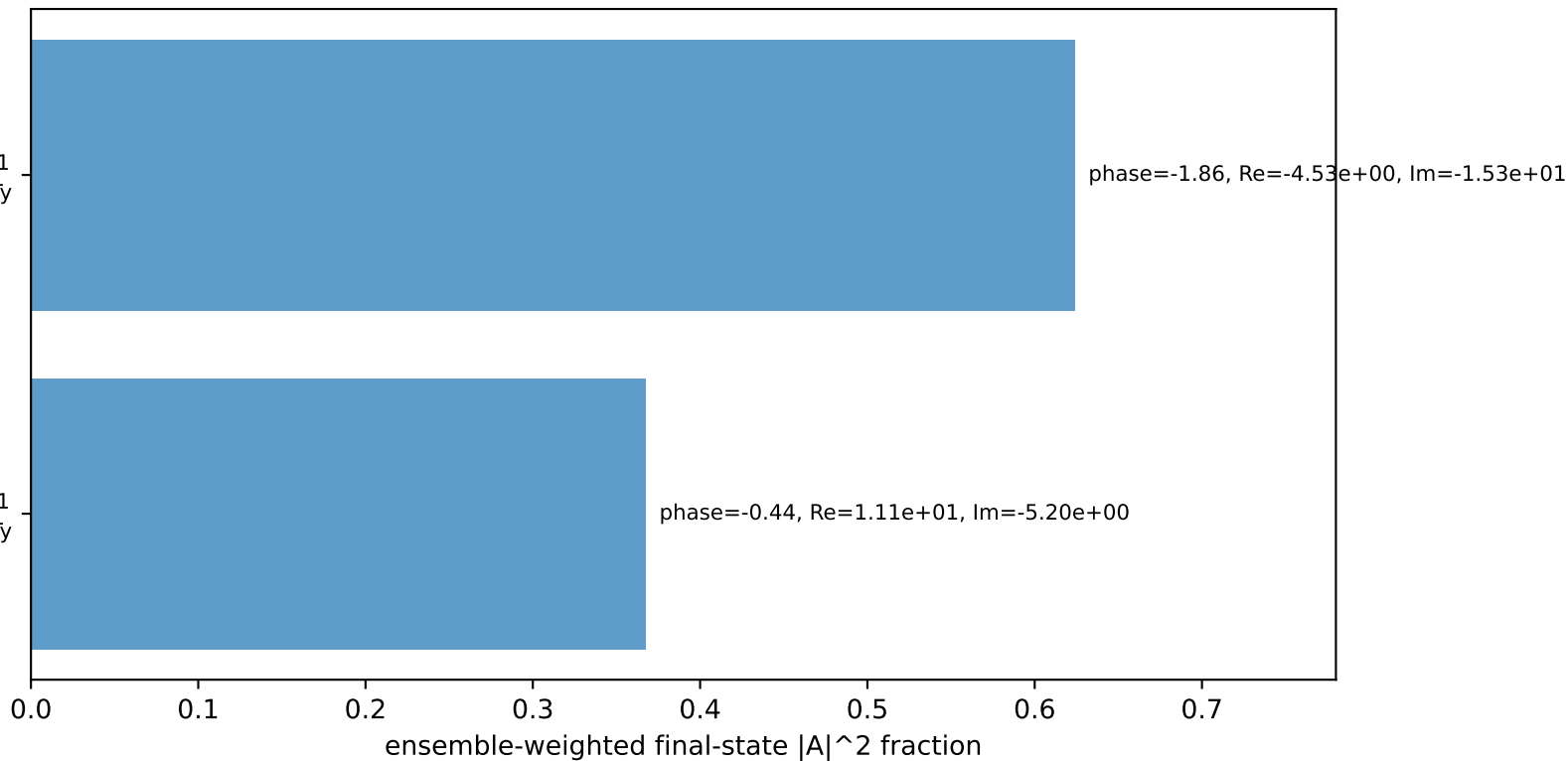
Transverse plane



$h_e=+1, h_p=-1, h_{\gamma}=+1$
 electron L, proton Ty

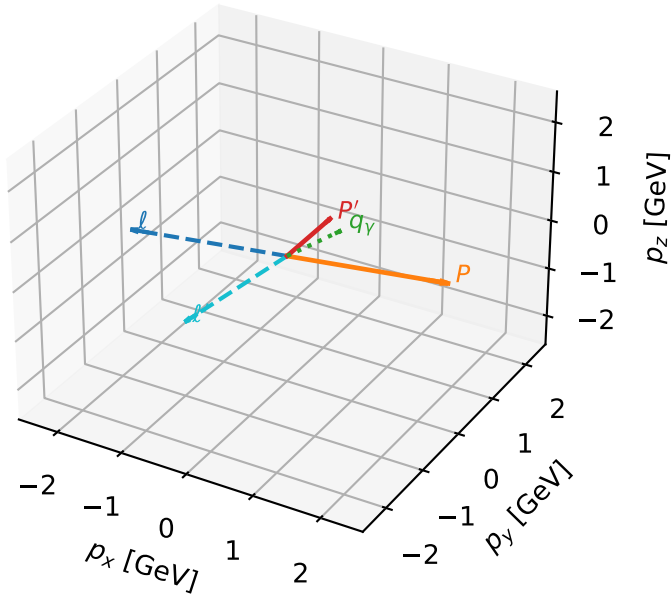
$h_e=+1, h_p=+1, h_{\gamma}=-1$
 electron L, proton Ty

Leading initial-ensemble/final-state components (N=2, retained=99.2%)



L massless lepton + Ty proton characteristic max F_3 configuration

3D momenta



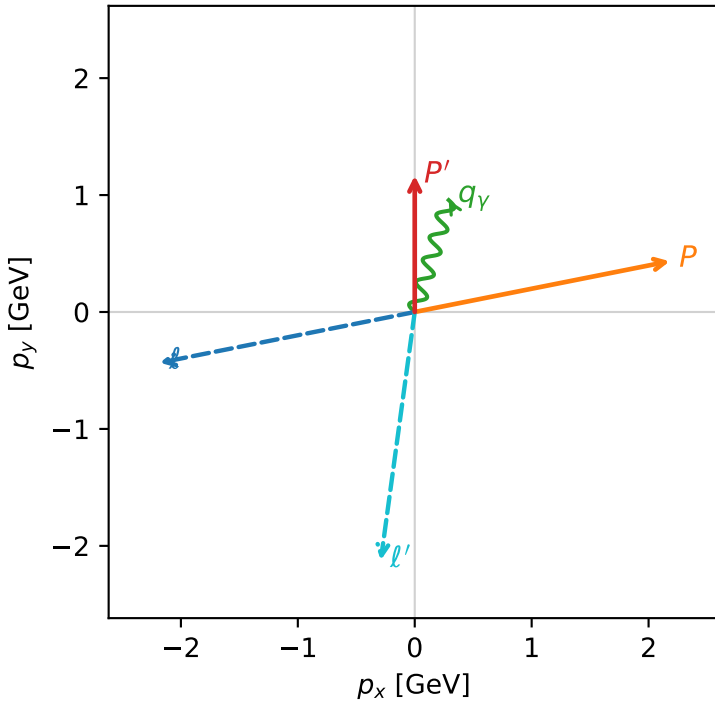
f3_mid_egamma_lty_region_2 F_3=2.40149e-08
 region: mid_Egamma_LTy_region_2
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|L_{\ell_0}\rangle \otimes |T_{Y,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=1.16564$
 $\theta_{in}=1.57079$, $\phi_{\ell}=3.33794$, $\phi_p=0.19635$
 $E_Y=1$, $\phi_Y=1.27627$
 $Q^2=6.42205$, $x_B=0.893998$, $t=-4.45249$, $W^2=1.64131$, $y=0.348106$
 $\theta(\ell', \gamma)=170.912$ deg

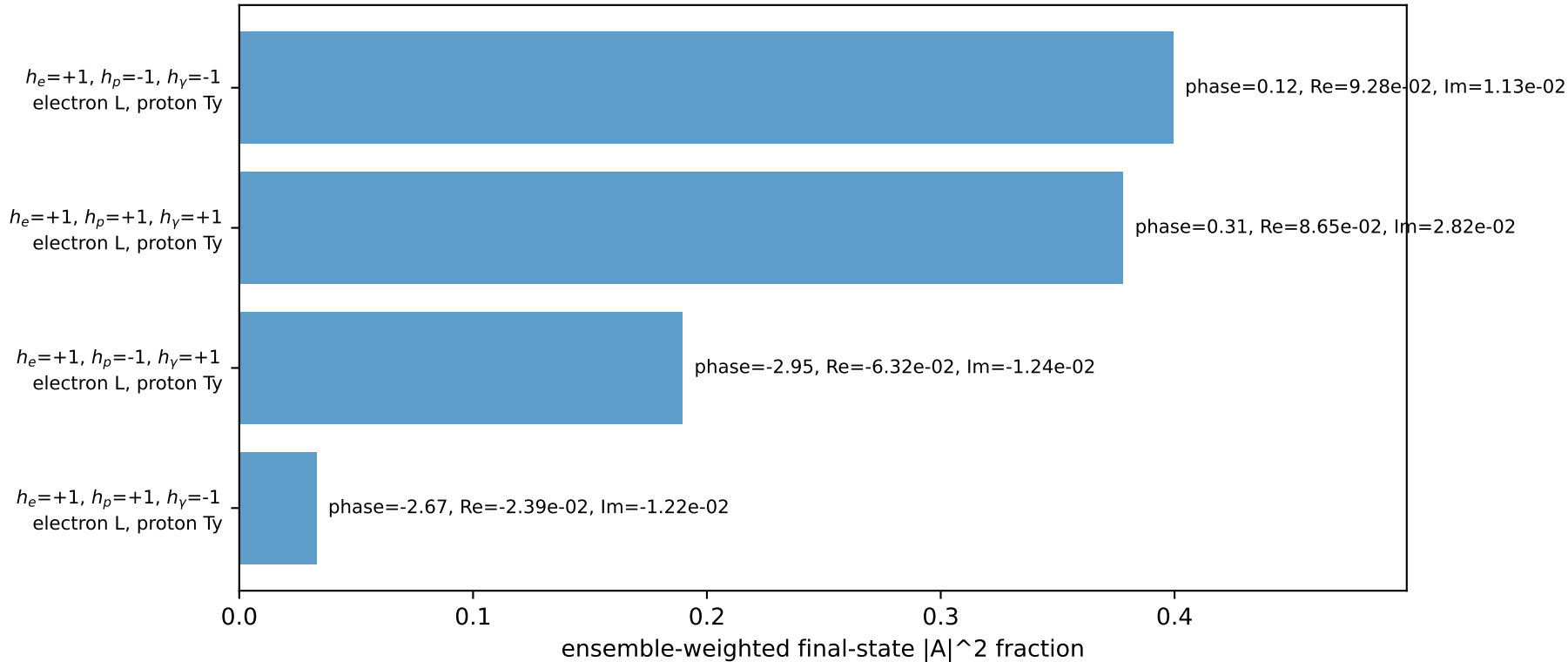
four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 2.2244$ $p_x= -2.1817$ $p_y= -0.4340$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= 2.1817$ $p_y= 0.4340$ $p_z= 0.0000$
 ℓ' $E= 2.1423$ $p_x= -0.2903$ $p_y= -2.1226$ $p_z= 0.0000$
 P' $E= 1.4962$ $p_x= 0.0000$ $p_y= 1.1656$ $p_z= 0.0000$
 q_Y $E= 1.0000$ $p_x= 0.2903$ $p_y= 0.9569$ $p_z= 0.0000$

Transverse plane

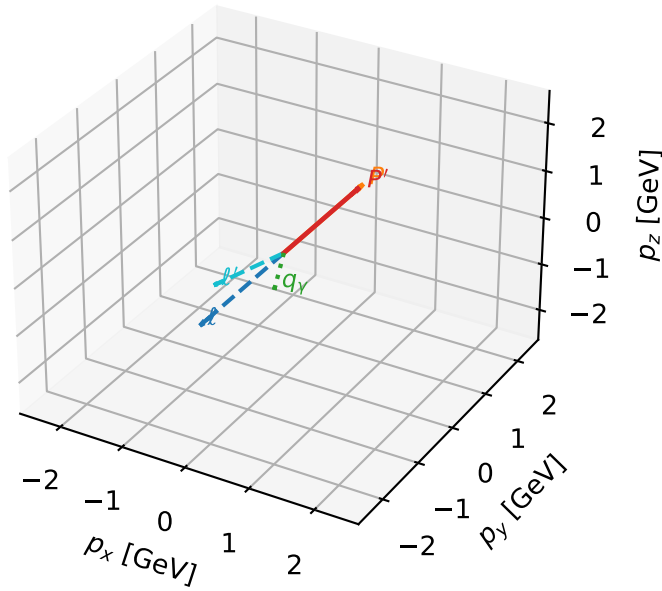


Leading initial-ensemble/final-state components (N=4, retained=100.0%)



L massless lepton + Ty proton characteristic max F_3 configuration

3D momenta

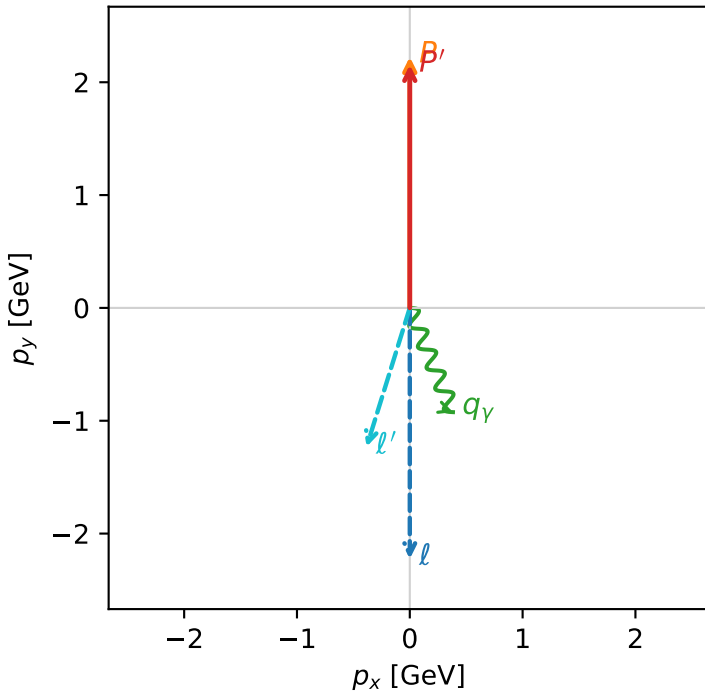


f3_mid_egamma_lty_region_3 F_3=2.04478e-08
 region: mid_Egamma_LTy_region_3
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|L_{\ell_0}\rangle \otimes |T_{Y,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=2.15447$
 $\theta_{in}=1.57079$, $\phi_{\ell}=4.71239$, $\phi_p=1.5708$
 $E_Y=1$, $\phi_Y=5.10509$
 $Q^2=0.25861$, $x_B=0.0289301$, $t=-0.000758984$, $W^2=9.56038$, $y=0.433182$
 $\theta(\ell', \gamma)=39.7745$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= -0.0000$ $p_y= -2.2244$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= 0.0000$ $p_y= 2.2244$ $p_z= 0.0000$
 ℓ' $E= 1.2887$ $p_x= -0.3827$ $p_y= -1.2306$ $p_z= 0.0000$
 P' $E= 2.3498$ $p_x= 0.0000$ $p_y= 2.1545$ $p_z= 0.0000$
 q_Y $E= 1.0000$ $p_x= 0.3827$ $p_y= -0.9239$ $p_z= 0.0000$

Transverse plane



$h_e=+1, h_p=-1, h_Y=+1$
 electron L, proton Ty

$h_e=+1, h_p=+1, h_Y=-1$
 electron L, proton Ty

$h_e=+1, h_p=-1, h_Y=-1$
 electron L, proton Ty

Leading initial-ensemble/final-state components (N=3, retained=100.0%)

